



Controls Academy – Plant Controls Overview

June 6, 2017

Presenters:

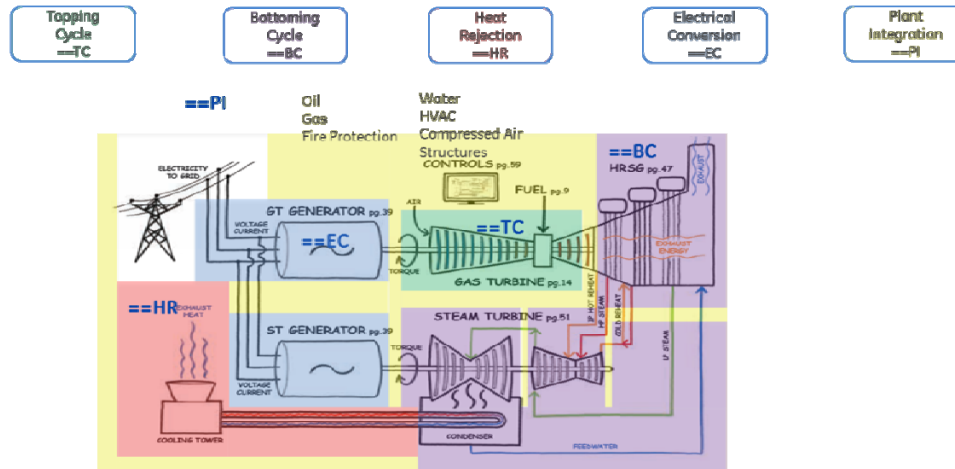
Gautam Chatterji

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What is Plant Controls?



Plant Controls:

Power gen equipment (TC, BC, EC)

+

BOP equipment

+

Plant perf & Emissions Guarantees

+

Operability guarantees

+

Integration

+

Control room



Plant Scope - Complexity / Variability:

- Integrated Solar (ISP) – Duba / WAS
- Integrated Heat (CHP) – Oseong / Anyang
- Integrated Gasification – Taeon; Inpex
- Integrated Desal (IWPP) – Marafiq
- Integrated Aero – Scattergood / Powerhouse
- Integrated Process – EMAL (Smelter) / Calc Pass (LNG)



Plant Controls – Distributed Nervous System of the Plant – Monitors, Protects, Operates & Communicates (On & Off-site)

Components of Plant Controls: Systems /Functions

Typical Combined-Cycle Plant Configuration

Topping Cycle	Bottoming Cycle	Heat Rejection	Electrical Conversion	Plant Integration	
Gas Turbine Gas Turbine Unit Cooling & Sealing Clearance Control Starting Means Steam Injection Gas Fuel Delivery Liquid Fuel Delivery LF Atomizing Air LF Water Injection GT Exhaust Bypass Stack GT Inlet Inlet Conditioning	Feedwater HRSG HP Steam IP Steam LP Steam Supplemental Firing SCR/ Ammonia Chemical Injection Blowdown Steam Steam Turbine Unit Steam Admission Steam Extraction Bypass ST Drains Steam Sampling Steam Seal Condensate Condensing Steam Condenser Vacuum Auxiliary Steam Deaerator / Sparging Pegging Steam	Circulating water Aux Cooling Water GT Cooling Water Gen Stator Cooling H2O	Generator Generator Unit Gas cooling Excitation Protection Electrical Evacuation Voltage Conversion High Voltage Auxiliary Power Distribution Medium Voltage Low Voltage Essential Services Electrical Starting Systems LCI Black Start Generation	Fuel Systems-Gas/Liquid Gas Conditioning Liquid oil conditions Oil System Hydraulic Oil / Lift Seal Oil Lube Oil Gas Systems H2, CO2, N2 Strg & Dist Fire Protection GT Fire Protection Plant Fire Protection ST Protection Water Systems DI Water System Raw Water GT Water Wash Potable Water Service Water Plant Drains Chilling	HVAC GT HVAC HVAC of GT Aux HVAC of ST Aux HVAC of Plant Aux Compressed Air Systems Service Air Instrument Air

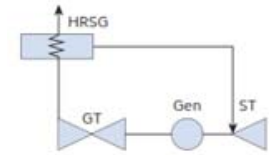
Integrated Plant Control System



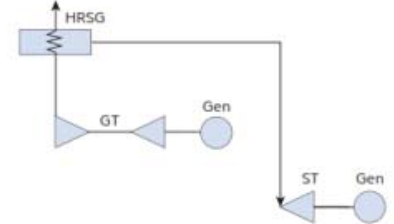
Combined Cycle Power Plant Configurations (CCPP)

Single-Shaft

- One single power train - GT/GEN/ST
 - Applies to E/F/H Class GT
 - D/A Class ST
 - Rigid coupling or clutched
 - Cascading or non-cascading bypass
 - Common auxiliaries

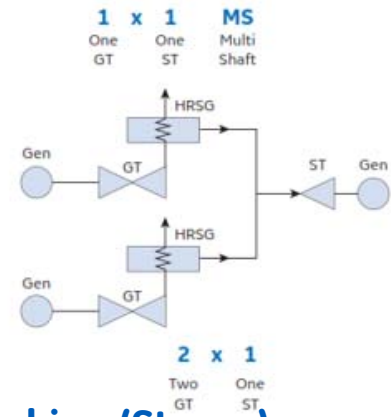


1 x 1 SS
One GT One ST Single Shaft



Multi-Shaft

- GT/GEN & ST/GEN on separate power trains
 - 1X, 2X and 3X configuration
 - Applies to E/F/H Class GT
 - D/A Class ST
 - Cascading or non-cascading bypass
 - With or without GT exhaust gas bypass damper



The Conventional CC couples the two thermodynamic cycles ... Brayton (Gas) & Rankine (Steam)

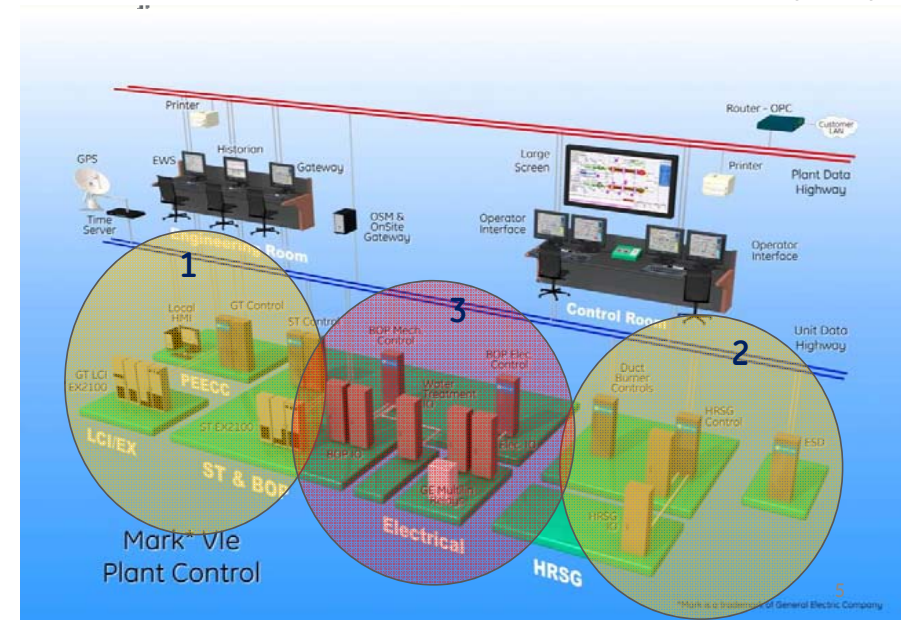
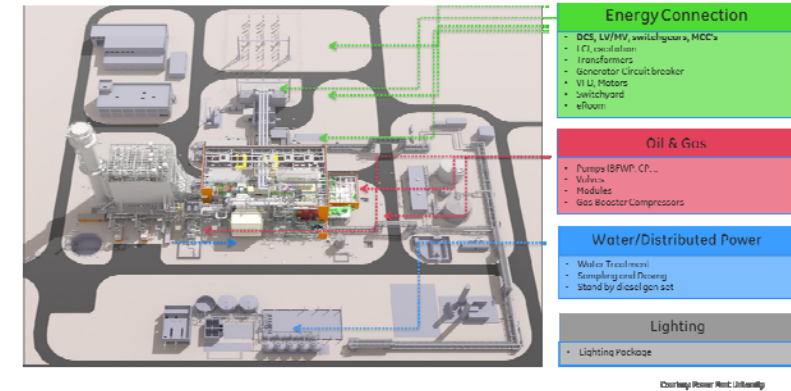


Project Structuring & Controls Scope

Business Model/ scopes of supply for GPS

Available scopes of supply

Gas Turbine and Generator Equipment	Combined Cycle Equipment	Engineered Equipment Package	Turnkey Power Plant
- Gas turbines including auxiliaries - Air inlet and exhaust system - Fuel gas system skids - GT generator including auxiliaries - Mark VIe controls for GT and generator - Fire protection - Options	- Steam turbine and auxiliaries without piping - Mark VIe controls for ST and generator - Options	- HRSG - Condenser - Fuel preheating and filtration - Options	Executed with EPC partner - Buildings/Civils - Plant cooling systems - Water treatment - Tanks - Plant piping/valves - Plant electrical equipment - Mark VIe plant controls - Erection/commissioning - Options
1	1	2&3	2&3



Why Plant Controls? Fulfilling Guarantees & Handling Transients

Contractual guarantees offered

- Output
- Emissions (inc. Transient)
- Noise
- Turndown
- Startup/Shutdown times
- Transient ramp rates

Examples:

Impact of Renewables on the Grid
Transient Emissions

Rapid response lite / full: GT-HRSG-BOP Interaction

Purge Credit: GT-HRSG-BOP Interaction

ST Trip: ST-HRSG-BOP Interaction

District Heating: Process-ST-HRSG Interaction

Integrated Solar impact: Process-HRSG Interaction

Our Promise... Guarantees

Emissions for Natural Gas & Distillate

Emissions	Units of Measure	Temperature Range	Gas Turbine Load	Gas Fuel Unfired	Gas Fuel Fired	Distillate Fuel Unfired
NO _x	ppmvd, Ref. 15%O ₂	0°F to 100°F	MECL to 100 % of Base Load	2	2	4
CO	ppmvd, Ref. 15%O ₂	0°F to 100°F	MECL to 100 % of Base Load	2	2	6
VOC as CH ₄	ppmvd, Ref. 15%O ₂	0°F to 100°F	MECL to 100 % of Base Load	1	2	3
NH ₃	ppmvd, Ref. 15%O ₂	0°F to 100°F	MECL to 100 % of Base Load	5	5	5
PM10/ PM2.5	lb/hr	0°F to 100°F	MECL to 100% of Base Load	14.4	22.6	75.6 (40% max)
Opacity	%	0°F to 100°F	MECL to 100% of Base Load	10	10	10

Near Field Noise

SPL, dBA	BS (Average)
Fuel	Natural Gas / Distillate Oil
Gas Turbine Load	Base

The average sound pressure levels (SPL) (re: 20 micropascals) from the indoor and/or outdoor supplier equipment defined in this technical specification, shall not exceed the value stated above, when measured 1 m (3 ft) in the horizontal plane and at an elevation of 1.5 m (5 ft) above the gas turbine operating level, steam turbine operating level (if different), and generator operating level (if different) identified on the General Arrangement drawings with each turbine generator set operating at the load stated above in accordance with contract specifications. Walkways and/or platforms that are not easily accessible by stairs are excluded from the above guarantee.

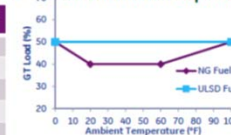
Areas under the steam turbine that are not considered operating levels (i.e., ground level and mezzanine level) are exempt from the above guarantee.

Near field guarantees apply to areas along a Site specific Source Envelope(s), determined by a line established 1 meter (3 ft) from the outermost surface of the equipment defined in the technical specification scope of supply including noise abatement equipment. Depending on the site arrangement and relationship of equipment locations, multiple source envelopes may be designated. (See sample Figure 3.3-1 below.)

Fuel Composition

Fuel Component	% Volume
CH ₄	94.726
C ₂ H ₆	4.569
C ₃ H ₈	0.212
C ₄ H ₁₀	0.032
C ₄ H ₁₂	0.007
C ₅ H ₁₂	0.001
CO ₂	0.156
N ₂	0.297
Total sulfur (ppm)	24
Total	100

GT Turndown Capacity



Startup Time

Event	Time (minutes)
Cold Start	175
Warm Start	95
Hot Start	51

Startup and Shutdown Emissions Duration - NG & ULSD Transient Ramp Rate

	Duration (minutes)
Cold Start	45
Warm Start	45
Hot Start	30
Shutdown	12

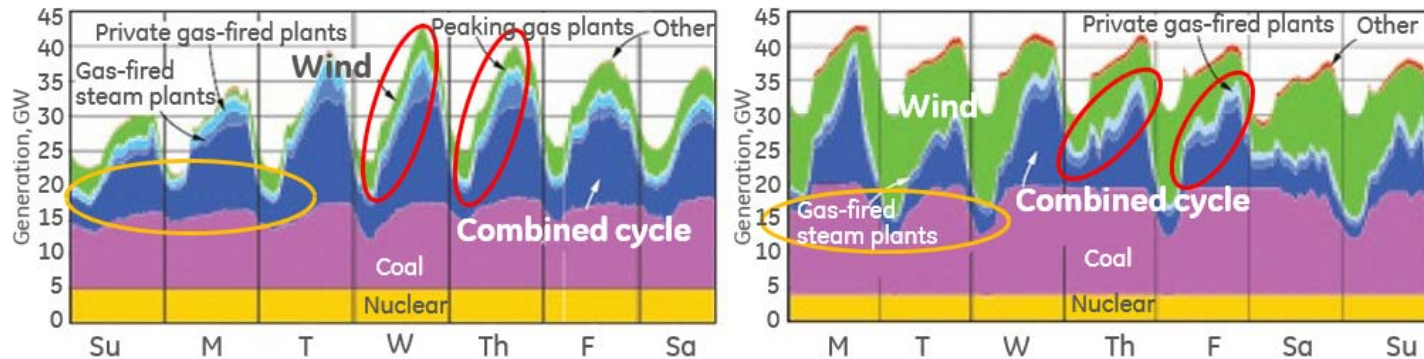
	Time (MW/min)
Ramp Rate	50

Integrated Plant Controls: Drives competitiveness / Enables GE to be more aggressive in guarantees for Start Time, Emissions, Plant Fast Maneuvering & Turndown... Reduces risk



Example: Impact of Renewables on the Grid

New Challenges for CCPP Operability



Prior to Renewables Penetration

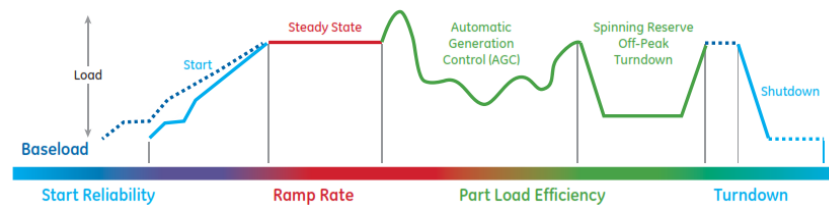
Nuclear/Coal

- Too cumbersome/slow

With Renewables Penetration

Renewables

- Variable



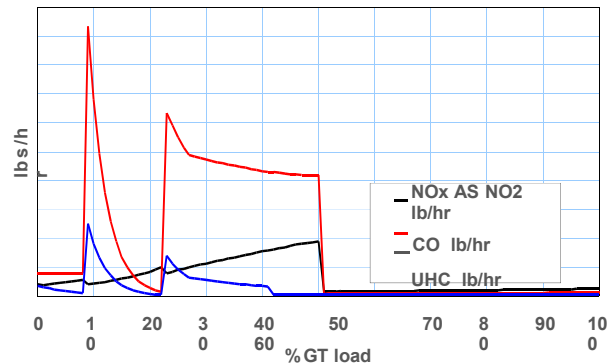
Combined Cycle Power Plant (CCPP) ensures the grid health by providing fast and reliable output due to it's flexibility and size



Example: Transient Emissions

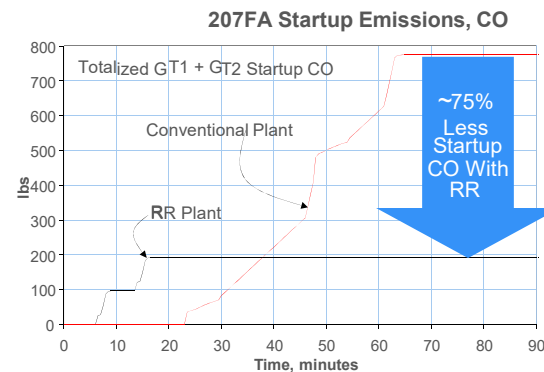
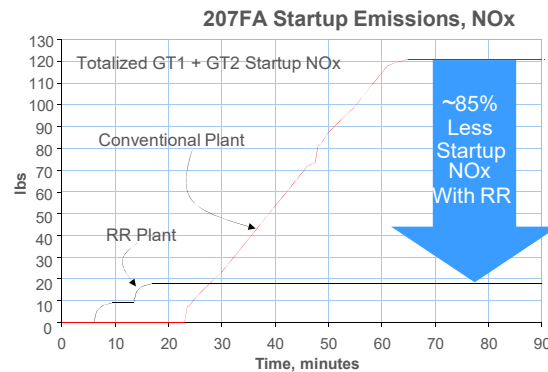
GE Rapid Response CC Plant - Benefits

Example DLN GT Emissions vs Load Characteristic



Emissions (lbs/start)

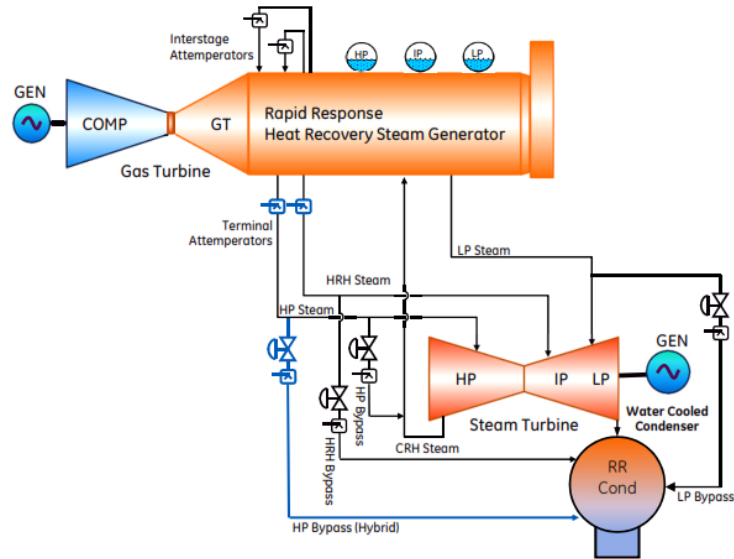
- Start/Stop emissions now being regulated on a “per start” basis, lbs per event, hot/warm/cold
- Reducing Start/Stop emissions can be critical to permitting
- New category of BACT for Start/Stop emissions



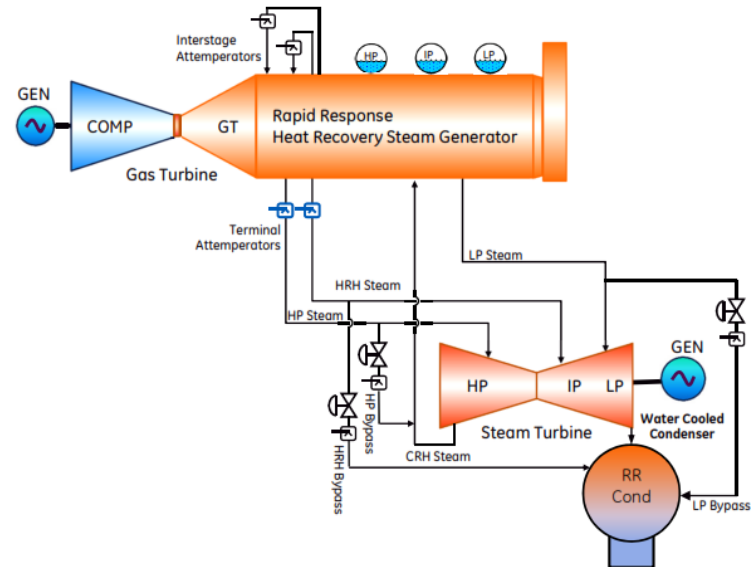
Fired / Unfired HRSG: Inlet NOx Measurement / NH3 Injection /SCR control

Example: Rapid Response Lite / Full - GT-HRSG-BOP Interaction

Power Plant Solutions



Rapid Response Plant

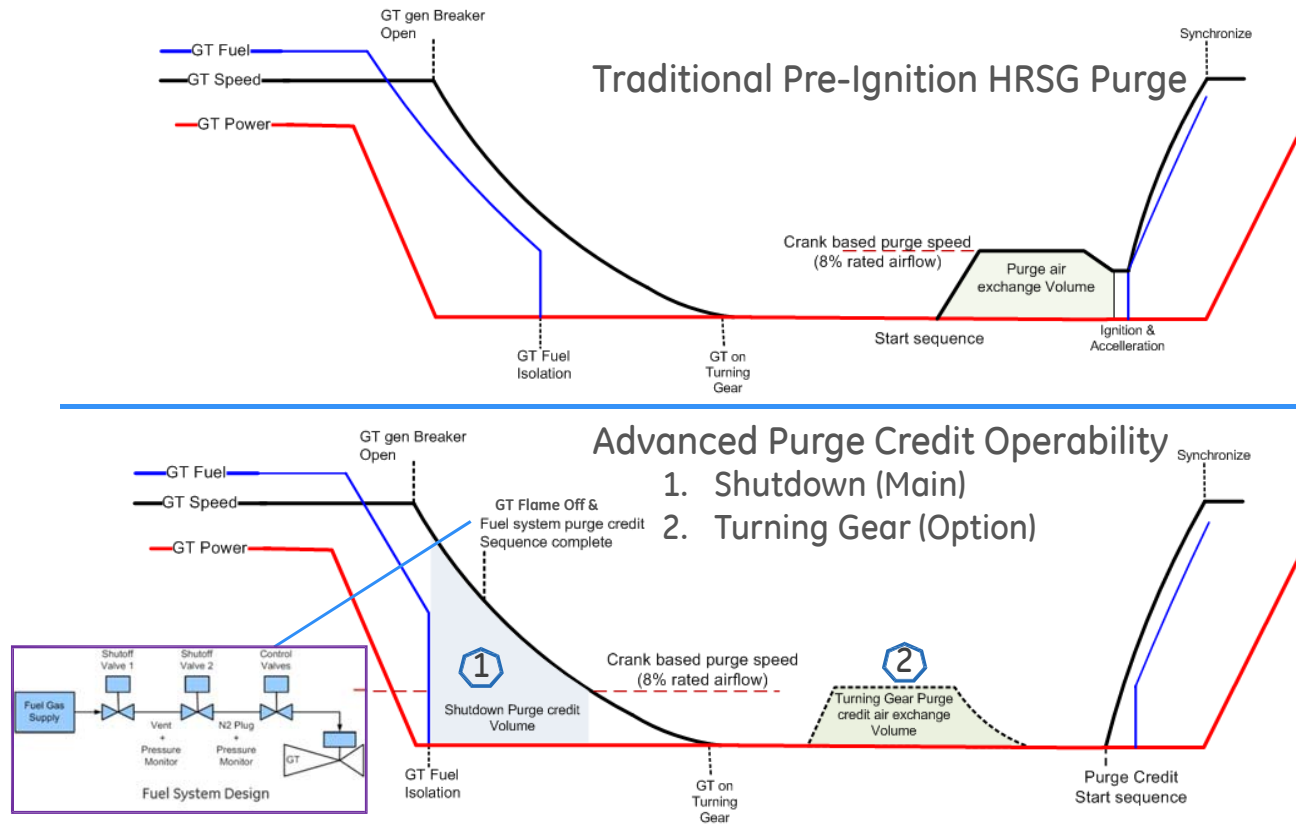


Rapid Response Lite Plant

Operability change for One component Impacts other systems



Example: Purge Credit - GT-HRSG-BOP Interaction



Interdependency of Systems – GT & HRSG Purge – cooldown – startup time

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Combined cycle System Transients

Examples:

ST Trip:

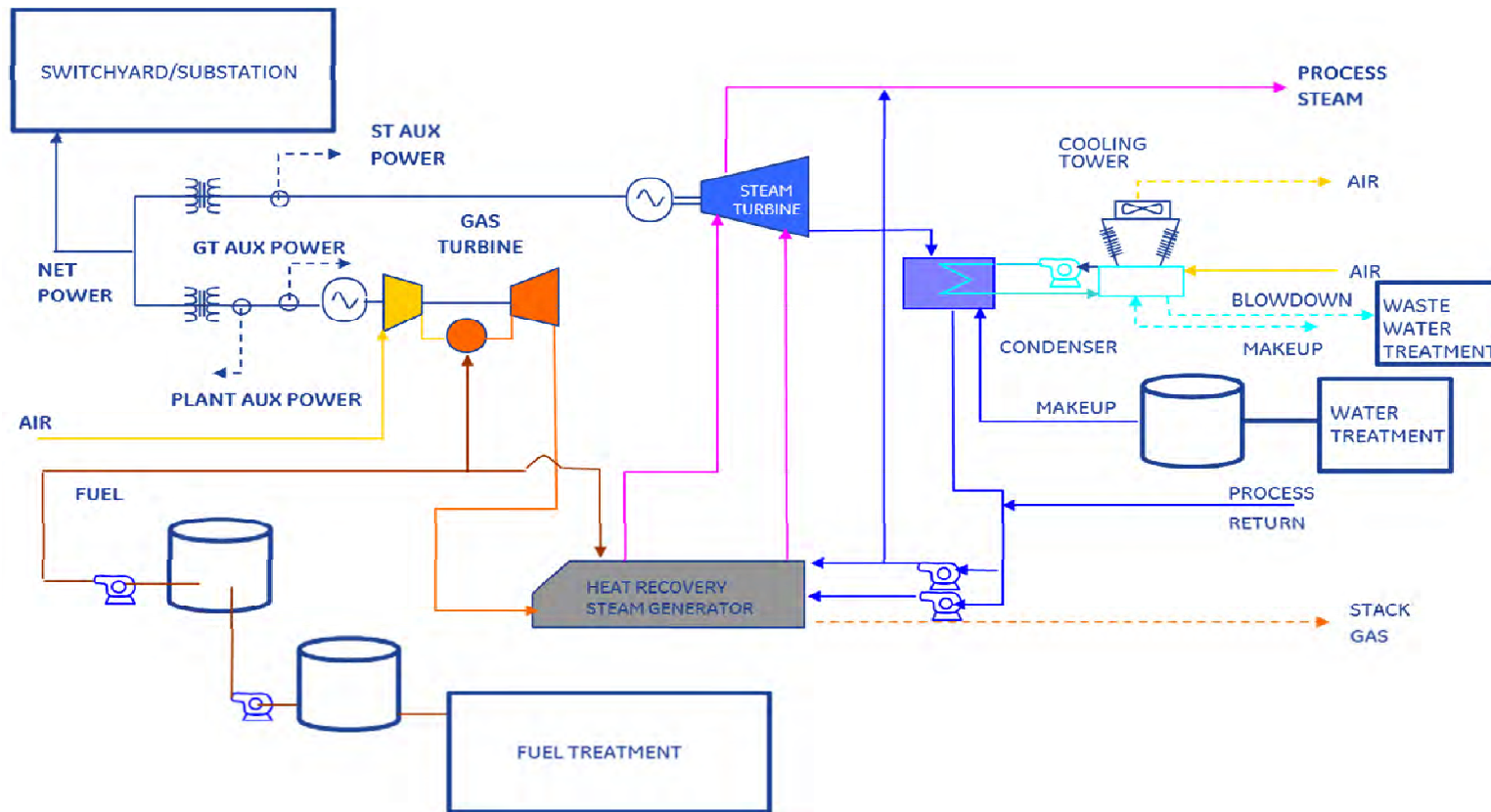
District Heating:

Integrated Solar impact:

ST-HRSG-BOP Interaction

Process-ST-HRSG Interaction

Process-HRSG Interaction



Interdependency of Systems - Plant level guarantees and transients



Plant Controls Stakeholders - Key drivers

EPC



- Optimized engg & commissioning cycle
- Seamless integration across plant
- Tools and access to data for reliability & performance tests
- Shorter Schedule and reduced cost

Plant Operator



- Operational effectiveness
- Better alarm mgmt
- Intuitive screen design
- Consistency across GT, ST, HRSG & BOP .. Entire plant
- Guidance for varied experience / expertise level

End Customer



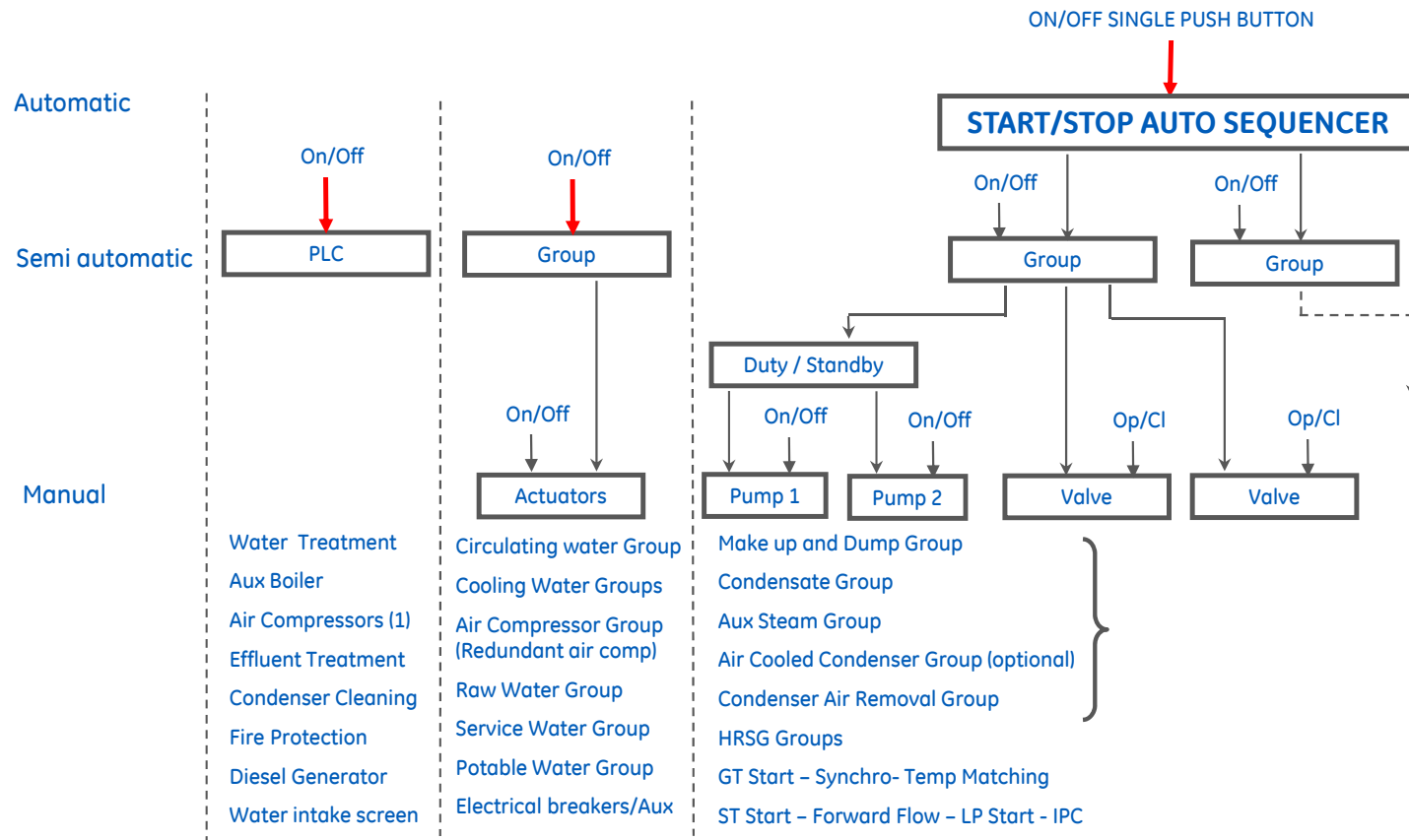
- Better asset management
- Predictable Performance - maximizing profitability
- Grid, Emission and Cyber Security compliance
- High Reliability and Availability
- Dispatch flexibility



Integrated Plant Controls catering to Different Requirements / expectations

Expectation: Level Of Automation – Integration is key

Higher Level of Automation: Start Stop Sequence supported by group control philosophy structure after Plant Pre Start conditions fulfilled



Operability Tests / Verification– Integration is key

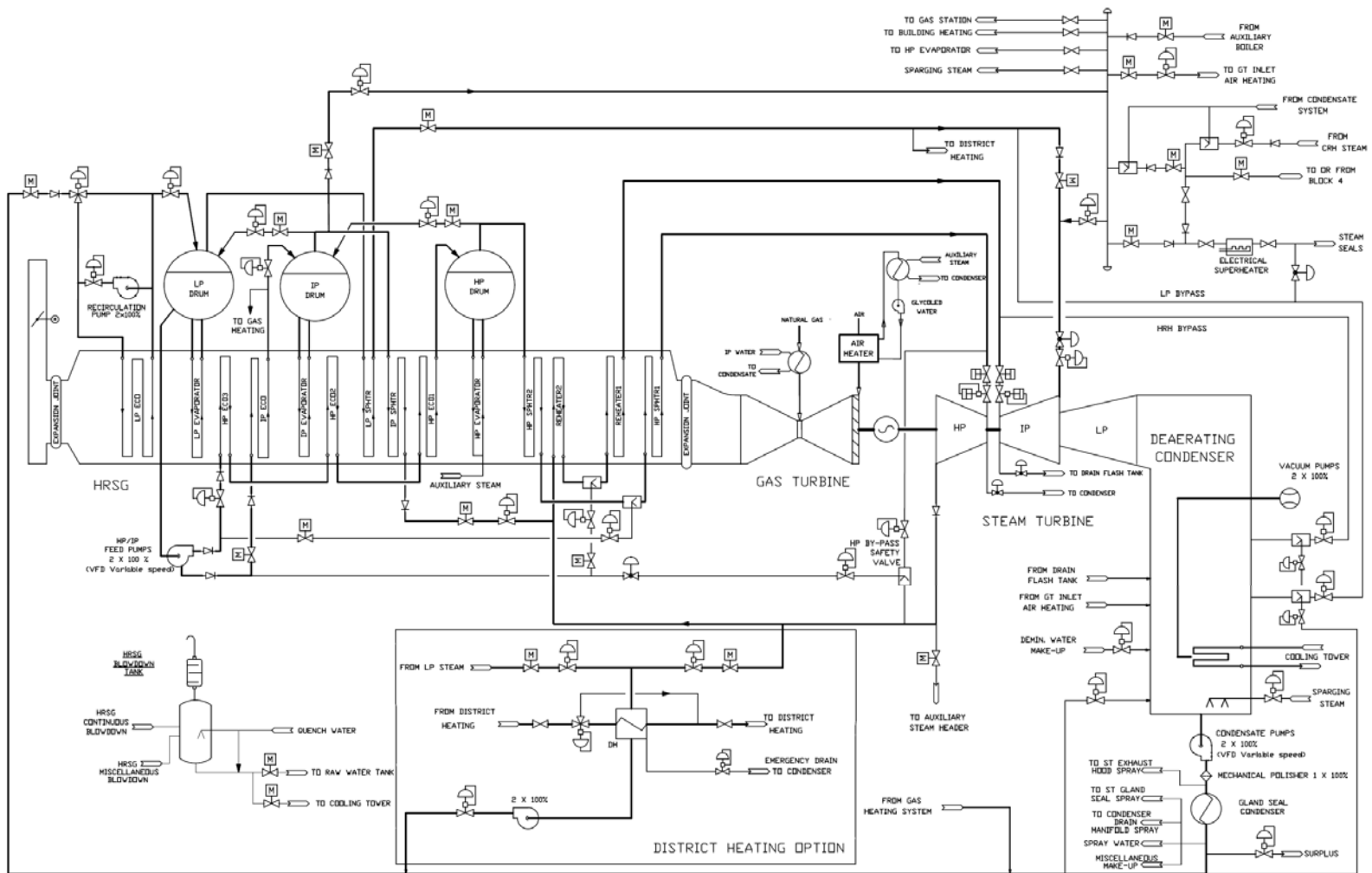
Video: <https://ge.box.com/s/37w7hex8r70d8bbtgz91lhs7uvflr0ns>

PLATFORM # 1 - RAPID RESPONSE PLANT									
RUN 1	CASE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	COLD	ISO	RUN 9	TYPE CONDITIONS CTQ CTQ	PLANT CONTINGENCY OPERATION PLANT TRIP / SHUTDOWN PURGE CREDIT / TURNING GEAR OPERATION RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO
RUN 2	CASE CONDITIONS CTQ CTQ	PLANT NORMAL OPERATION PLANT LOOP TUNING / PLANT MIN TURNDOWN / FAST RAMP RATE RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO	RUN 10	TYPE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE STARTUP TIME - BASELINE DEFINITION CONTROLS VALIDATION (EXPECTED TO BE AUTO/MANUAL)	HOT	ISO
RUN 3	CASE CONDITIONS CTQ CTQ	PLANT SHUTDOWN NORMAL SHUTDOWN PURGE CREDIT / TURNING GEAR OPERATION RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO	RUN 11	TYPE CONDITIONS CTQ CTQ	PLANT CONTINGENCY OPERATION PLANT ISLANDING OPERATION (15% GT LOAD) RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO
RUN 4	CASE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	HOT	ISO	RUN 12	CASE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE LITE OPERATING SEQUENCE CONTROLS VALIDATION	COLD	ISO
RUN 5	CASE CONDITIONS CTQ CTQ	PLANT CONTINGENCY OPERATION HRSG PUMP SWITCH-OVER / PLANT FULL LOAD REJECTION RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO	RUN 13	CASE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE LITE OPERATING SEQUENCE CONTROLS VALIDATION	HOT	ISO
RUN 6	TYPE CONDITIONS CTQ CTQ	PLANT STARTUP RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	WARM	ISO	RUN 14	CASE CONDITIONS CTQ CTQ	PLANT STARTUP CONVENTIONAL OPERATING SEQUENCE CONTROLS VALIDATION	COLD	ISO
RUN 7	TYPE CONDITIONS CTQ CTQ	PLANT CONTINGENCY OPERATION PLANT LOAD REJECTION TO SPINNING RESERVE RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO	RUN 15	CASE CONDITIONS CTQ CTQ	PLANT STARTUP CONVENTIONAL OPERATING SEQUENCE CONTROLS VALIDATION	HOT	ISO
RUN 8	TYPE CONDITIONS CTQ CTQ	PLANT STARTUP STEAM TURBINE START-ON-THE-FLY RAPID RESPONSE OPERATING SEQUENCE CONTROLS VALIDATION	HOT	ISO	RUN 16	CASE CONDITIONS CTQ CTQ	PLANT SHUTDOWN NORMAL SHUTDOWN PURGE CREDIT / TURNING GEAR OPERATION CONVENTIONAL OPERATING SEQUENCE CONTROLS VALIDATION	N/A	ISO



Integrated Plant Controls: Imperative for reducing Plant Guarantee & Customer risk

System split over the bottoming cycle



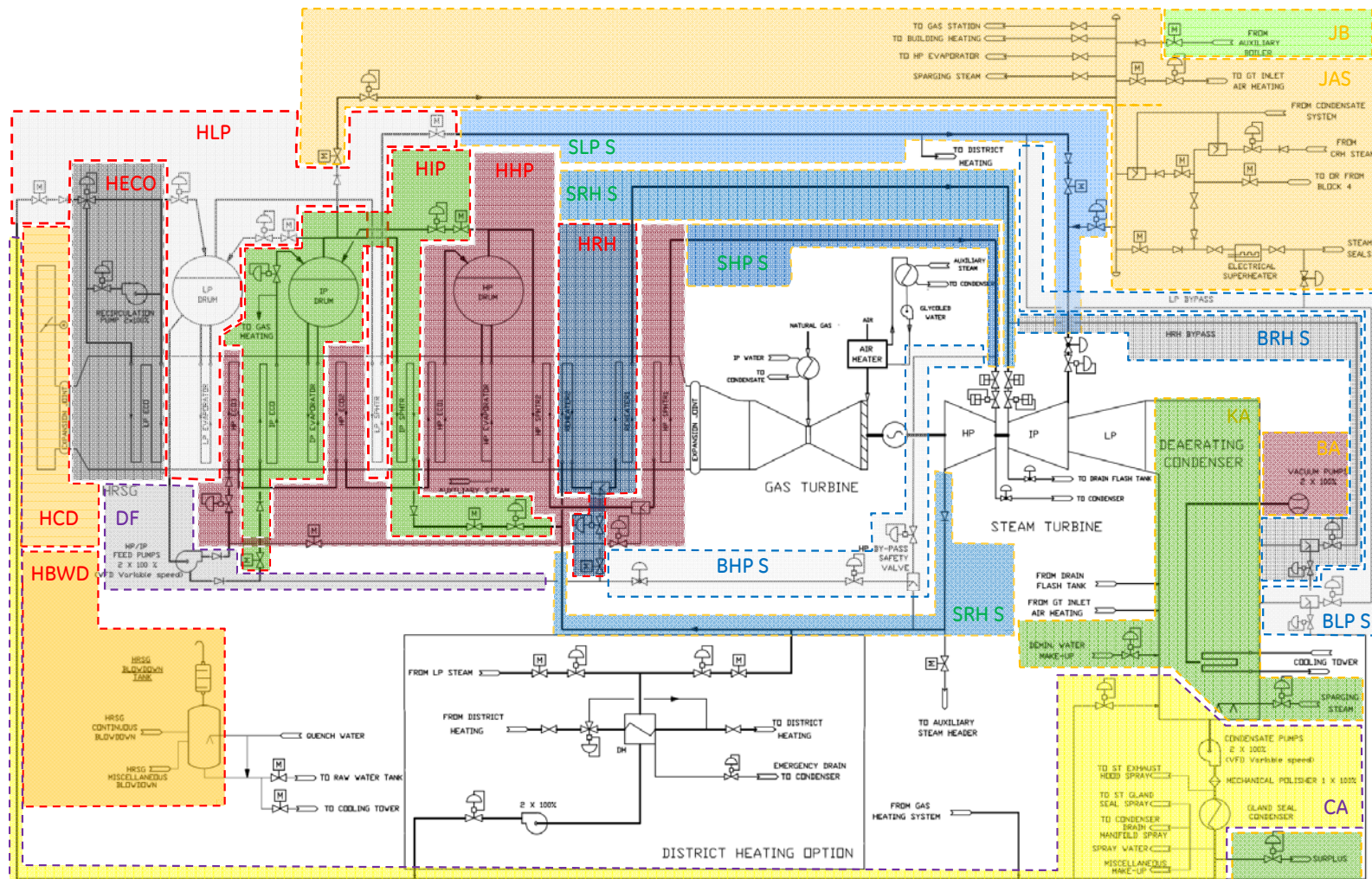
Design Approach: How is Controls Designed for a Complex Process

To better control and operate the plant:

- Split in systems
 - 1 system = 1 group = 1 elemental function,
- Independant HW,
- Independent SW,
- Interacts with others,
- To gear in the right way !!! -> Autosequencer
- To coordinate GT, ST, HRSG, BOP -> Plant load control, Steam flow & pressure control, Rapid response, HRSG protections.....

>> This make integrated plant





Hardware Design - Cycle

Hw Design cycle:

Process scope → group scope → Device list → IO's/device → IO list

= P&ID's = T201 = Dev.list/group = CIP

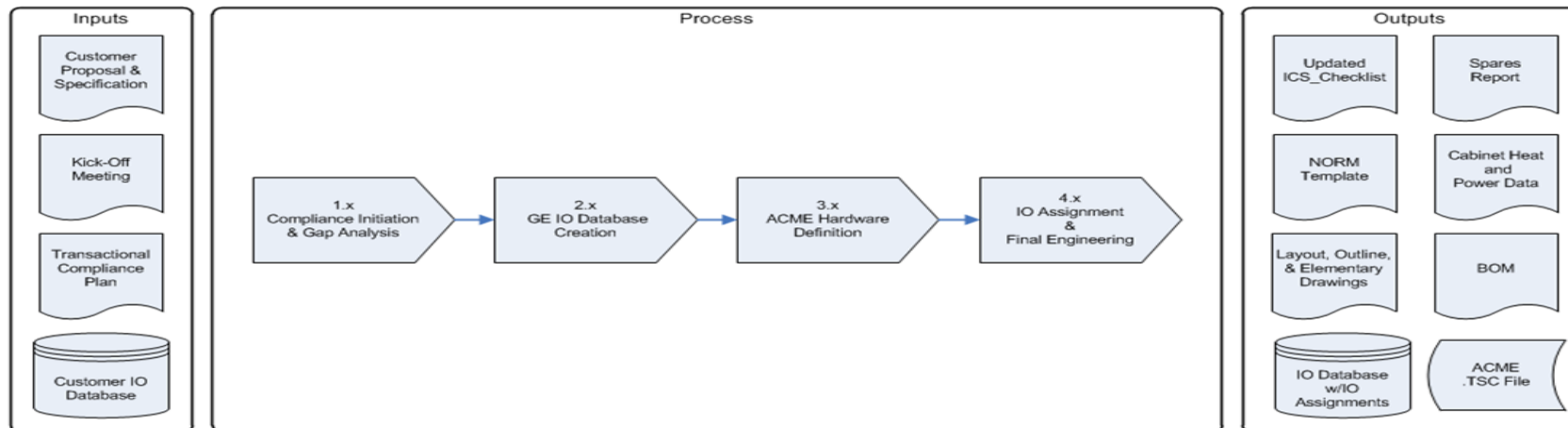
→ panel Layout → (CO0) Manufacturing → Drawings & BOM → Testing → Site

Rule 1: group definition

plant split in groups or systems, cf System Engineering

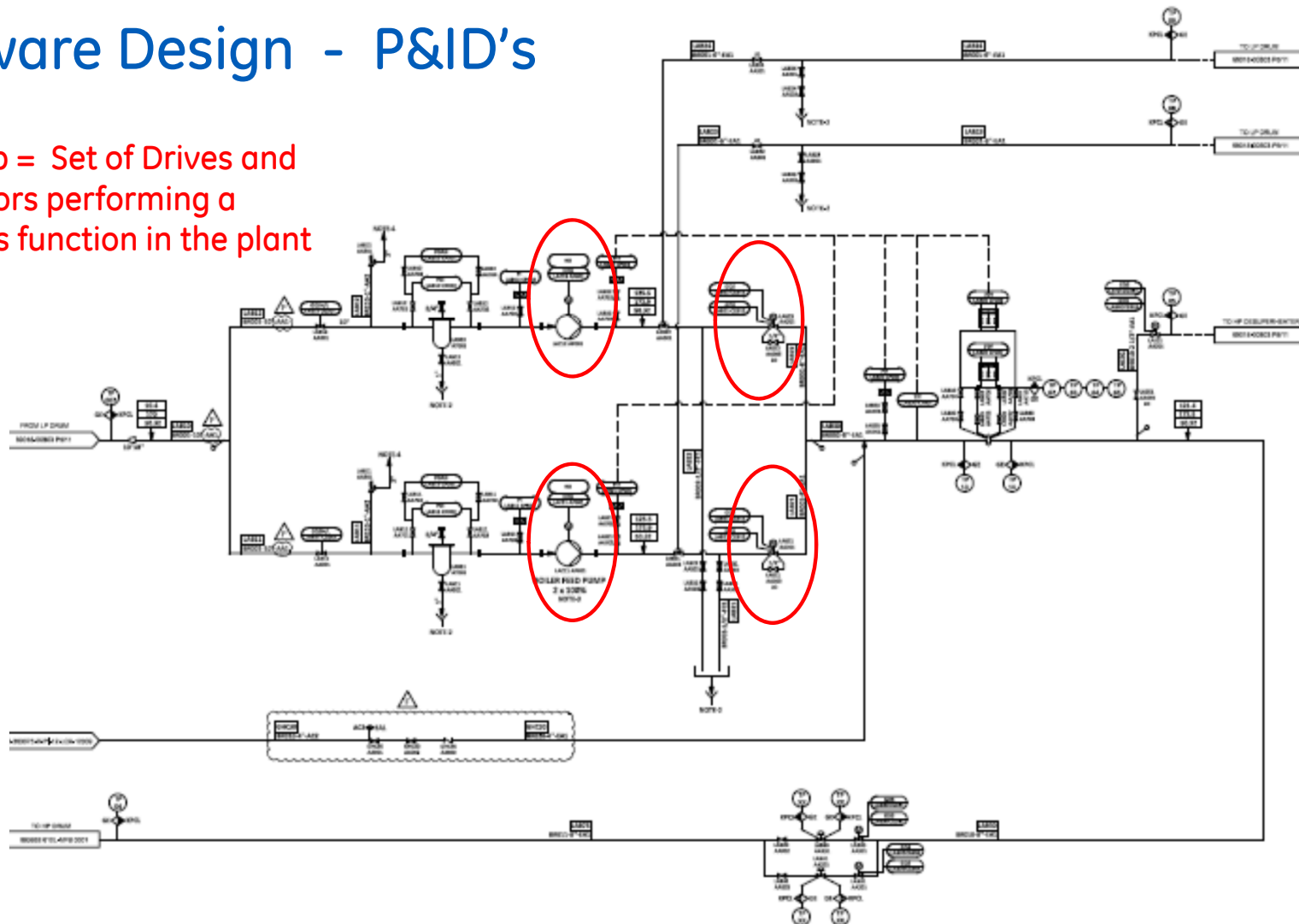
Plant = m x functional groups = n x Controllers

>> groups are distributed in the controllers



Hardware Design - P&ID's

1 group = Set of Drives and actuators performing a process function in the plant



Hardware Design - T201 extract

1.1 HP Feedwater Pumps Group

Scope of control: **KPCL**

P&ID Owner: **KPCL (Feedwater System P&ID: MM-363075-M-PI-12-LAB-12008)**

Logic controller: **HRSG Controller (PCS)**

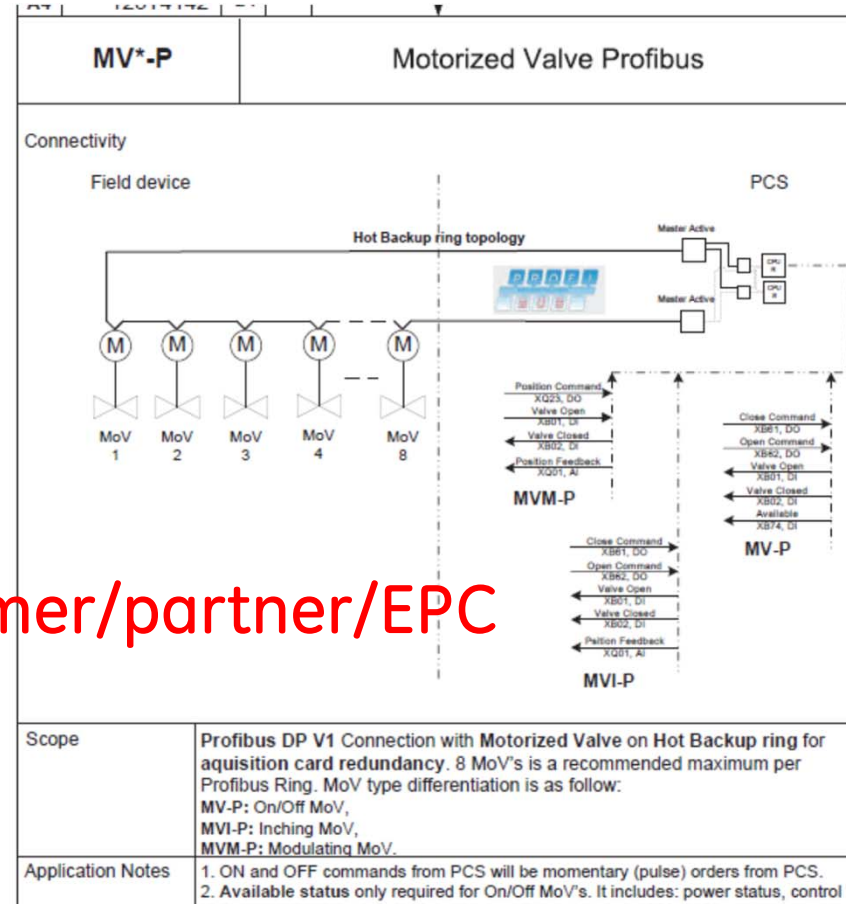
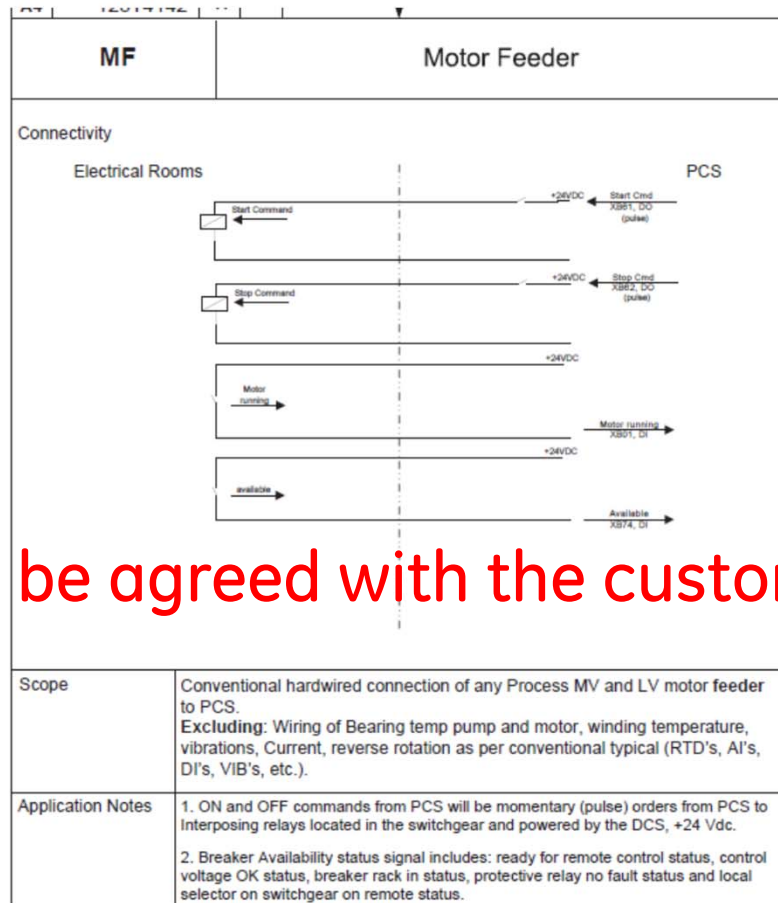
Start-up/Shutdown order initiated by the Auto sequencer: **YES**

System Description reference (KPCL): **MM-363075-M-DC-12-L-12301**

Equipment Description	KKS Tag	PID
HP feedwater pump 1	LAC11AP001	MM-363075-M-PI-12-LAB-12008 revH
HP Feedwater pump 1 discharge isolation valve	LAB21AA301	MM-363075-M-PI-12-LAB-12008 revH
HP feedwater pump 2	LAC12AP001	MM-363075-M-PI-12-LAB-12008 revH
HP Feedwater pump 2 discharge isolation valve	LAB22AA301	MM-363075-M-PI-12-LAB-12008 revH



Hardware Design - CIP



Hardware Design - I/O List

DEV_DEF

OWNER	MOD	Scope	FLD_DEV_TYPE	FLD_DEV_TY PE_PT	FLD_DEV_DESC	LVL3	DESC_FUNC	DESC_FUNC_S	LVL2	SIG_TYPE	SIG_LEVEL	PWR_SRC
KPCL		Mech	PVM1_a	1	MOV O/C	XB01	OPEN	OPEN		DI	24 VDC	S
KPCL		Mech	PVM1_a	2	MOV O/C	XB02	CLOSED	CLOSED		DI	24 VDC	S
KPCL		Mech	PVM1_a	3	MOV O/C	XB20	LOCAL CTRL	LOCAL CTRL		DI	24 VDC	S
KPCL		Mech	PVM1_a	4	MOV O/C	XB61	OPEN ORD	OPEN ORD		DO	24 VDC	S
KPCL		Mech	PVM1_a	5	MOV O/C	XB62	CLOSE ORD	CLOSE ORD		DO	24 VDC	S
KPCL		Mech	PVM1_a	6	MOV O/C	XB74	AVAILABLE	AVAILABLE		DI	24 VDC	S
KPCL		Mech	PVM1_a	7	MOV O/C	XB88	TORQ SW CL	TORQ SW CL		DI	24 VDC	S
KPCL		Mech	PVM1_a	8	MOV O/C	XB87	TORQ SW OP	TORQ SW OP		DI	24 VDC	S

DEV LIST

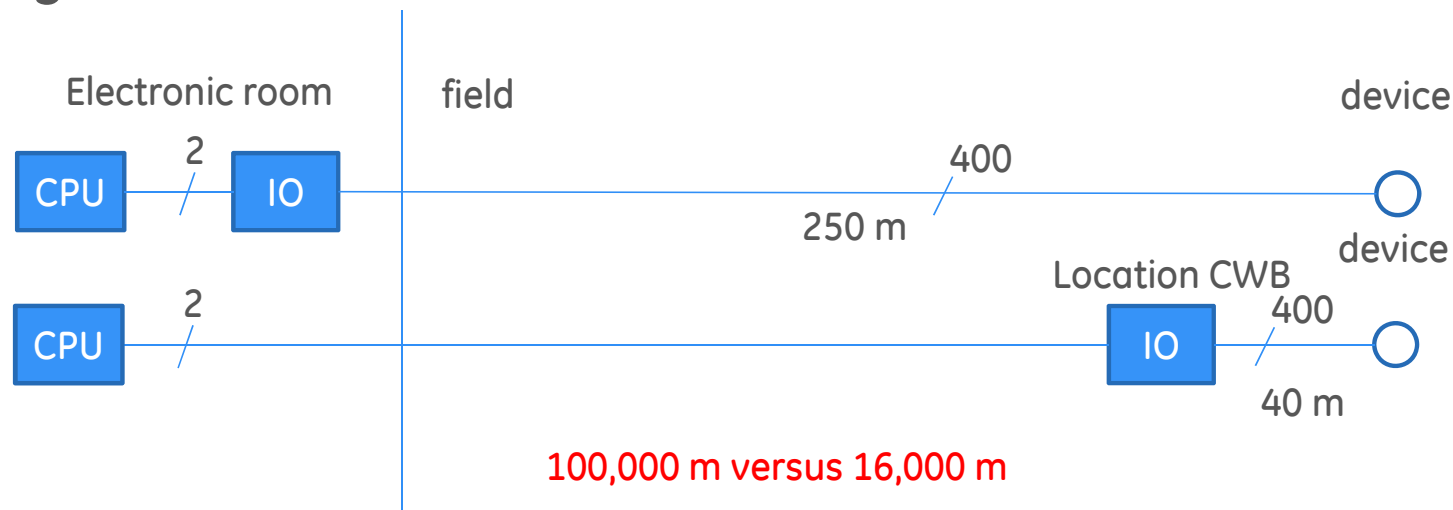
OWNER	MOD	FLD_DEV_T YPE	FLD_DEV	LVL1	LVL2	DESC_MAIN	DESC_MAIN_S	DEV_PART ITION	SYSTEM	CONTROLLER	LOCATION
GE		AI01	HAC20CF001A	HAC20	CF001A	LP FW FLOW	LP FW FLOW	A	Plant.LP_DRUM	H1	CWPH
GE		AI01	HAC20CF001B	HAC20	CF001B	LP FW FLOW	LP FW FLOW	B	Plant.LP_DRUM	H1	CWPH
GE		SW_a	HAD20CG301	HAD20	CG301	HP FWP1 MIN FLOW IV OPEN	HP FWP1 MIN FLOW IV OPEN		Plant.FWP	H1	EER
GE		SW_a	HAD20CG303	HAD20	CG303	HP FWP2 MIN FLOW IV OPEN	HP FWP2 MIN FLOW IV OPEN		Plant.FWP	H1	EER
GE		AI04	LAB75CT001	LAB75	CT001	HP FW TEMP	HP FW TEMP		Plant.HP_DRUM	H1	CWPH
KPCL		AI01	LAB30CP001	LAB30	CP001	HP FWP OUT PRES	HP FWP OUT PRES		Plant.FWP	H1	EER
KPCL		AI01	LAB30CT001	LAB30	CT001	HP FWP OUT TEMP	HP FWP OUT TEMP		Plant.FWP	H1	EER
KPCL		SW_a	LAB11CG301	LAB11	CG301	HP FWP1 INL IV OPEN	HP FWP1 INL IV OPEN		Plant.FWP	H1	EER
KPCL		AI01	LAB11CP301	LAB11	CP301	HP FWP1 FLT DP	HP FWP1 FLT DP		Plant.FWP	H1	EER
KPCL		SW_a	LAB12CG301	LAB12	CG301	HP FWP2 INL IV OPEN	HP FWP2 INL IV OPEN		Plant.FWP	H1	EER
KPCL		AI01	LAB12CP301	LAB12	CP301	HP FWP2 FLT DP	HP FWP2 FLT DP		Plant.FWP	H1	EER
KPCL		PVM1_a	LAB21AA301	LAB21	AA301	HP FWP1 OUT IV	HP FWP1 OUT IV	A	Plant.FWP	H1	EER
KPCL		AI01	LAB21CP001	LAB21	CP001	HP FWP1 OUT PRES	HP FWP1 OUT PRES	A	Plant.FWP	H1	EER
KPCL		PVM1_a	LAB22AA301	LAB22	AA301	HP FWP2 OUT IV	HP FWP2 OUT IV	B	Plant.FWP	H1	EER
KPCL		AI01	LAB22CP001	LAB22	CP001	HP FWP2 OUT PRES	HP FWP2 OUT PRES	B	Plant.FWP	H1	EER



Hardware Design - Plant Lay Out

Rule 2: Localization

IO's panel can be placed close to field equipment in order to save cabling



Hardware Design - Plant Lay Out

Rule 3: Controller sizing

1 controller = around 1000 IO's max, around 50% CPU Load

>> to be multiplied as required

+ Software complexity → + CPU Load → + Controller

In a Multishaft Plant:

1/GT; 1/ST; 1/HRSG; 1/BOPM/HSRG; 1/BOPM/Common; 1/BOPE



Control Room - Integrating the plant



Software Design - Cycle

Process scope → Group scope → Group operation → Coding →
= P&ID's = T201 = T202's + T207 = Software
Narratives & settings

Simulation → Eng.Validation → Integrated SWAT → Site
= Model = Test = Cust.test = Commissioning

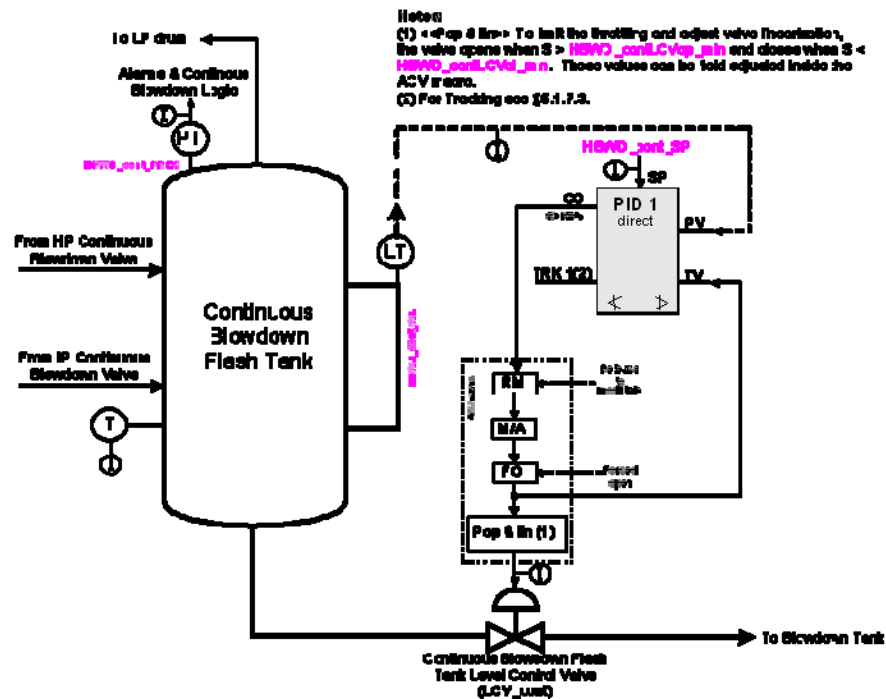
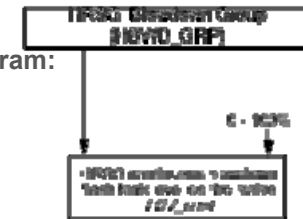


Software Design - Narrative

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- [6.0 Open / Closed Valve Sequences](#)
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HRSG Blowdown Group Organization Diagram:



Software Design - Narrative

5 Control Valve Sequences

Continuous Blowdown Flash Tank Level Control Valve:

Description:

One continuous blowdown flash tank that receives the outflow from the HP continuous blowdown drains. From this tank the steam part goes to the LP drum and the water flows to the blowdown tank.

References:

HRSG HP Group

Operation:

Normal Combined Cycle Operating Case:

Manual Action: This allows the operator to assume manual control of the valve and position it from 0% to 100% if there is no forced open condition.

Auto Action: The continuous blowdown flash tank is in service when the HRSG Blowdown Group is "ON". In the flash tank the blowdown separates into steam and water with drum blowdown constituents; the steam goes to the LP drum, and the water with drum blowdown constituents goes into the HRSG blowdown tank. A control valve maintains the level of the flash tank at its setpoint value **HBWD_cont_SP**.

Setpoint Action: The setpoint is a fixed value **HBWD_cont_SP**.

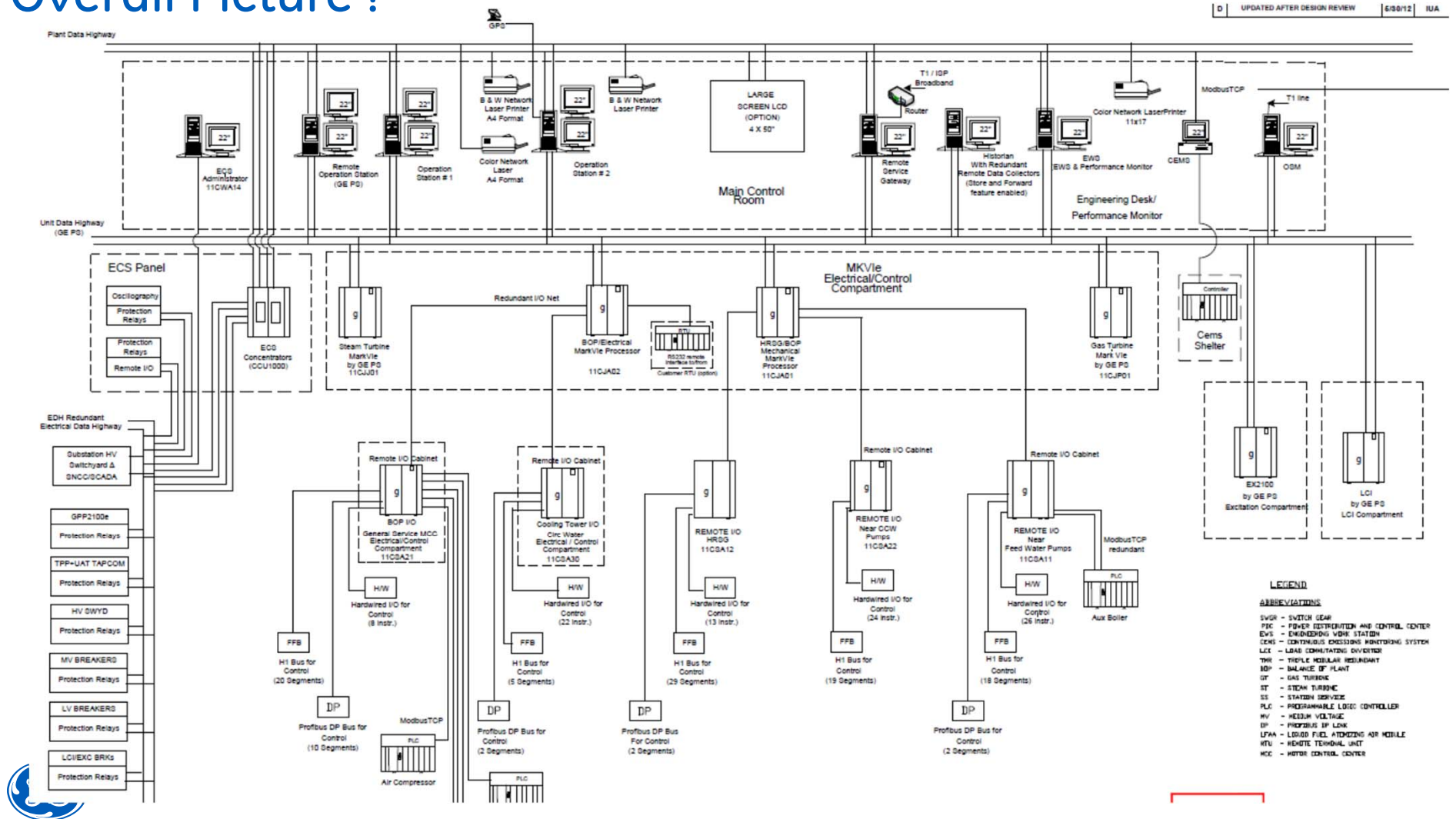
Equipment Protection:

etc.....

Include:
Permissives list
Table of Control Parameters
Alarm list



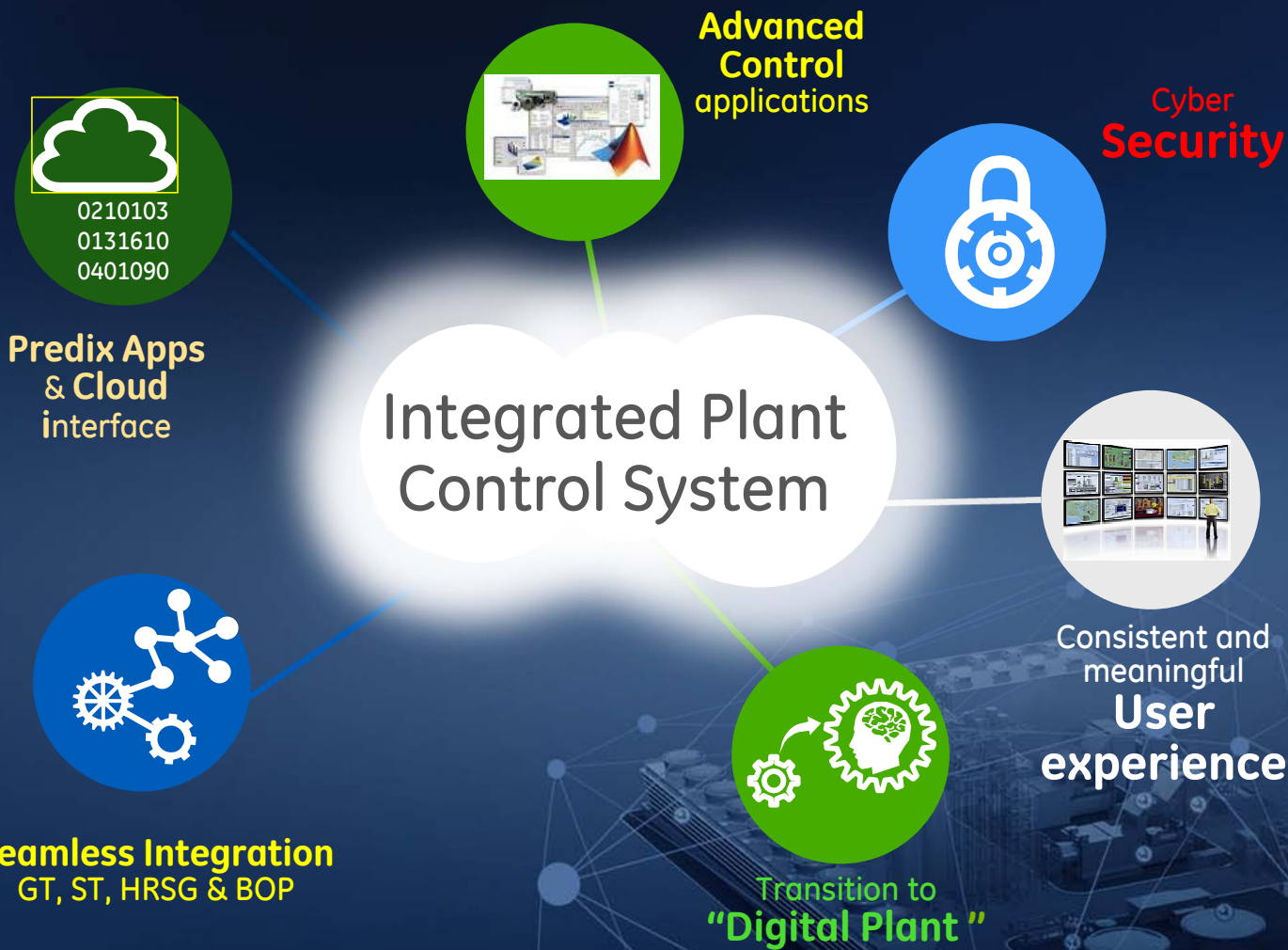
Overall Picture !



Controls Academy – Upcoming Sessions

- Plant Controls Systems A Deeper Dive !
- Plant Operability
- Systems thinking
- HRSG Systems





Class II - GE Internal